Gandaki University (2025: Mock Test)

Faculty of Science & Technology — **Full Marks: 50**

Bachelor's Degree (BIT) – Semester II — **Pass Marks: 25**

Course Title: Web Technology — **Time: 2 Hours**

Write your answers clearly and neatly. Use examples and diagrams where necessary.

Section A – Short Answer Questions (5 × 2 = 10 Marks)

(Attempt all questions)

1. Define HTTP and HTTPS. What is the difference between them?
2. Write the full form of FTP and POP. State their basic function.
3. What is the purpose of <colspan> and <rowspan> in HTML tables?
4. Mention any two advantages of using CSS in website design.
5. Write a basic JavaScript code to display an alert message saying "Welcome to Web Tech!"

Section B – Descriptive Questions (5 × 5 = 25 Marks)

(Attempt any 5 questions)

6. Explain different types of web hosting with suitable examples.
7. Write HTML code to create a form that collects user name, gender, age, and password with appropriate input types.
8. Define and demonstrate three types of CSS selectors with syntax and example.
9. Describe audio and video integration in HTML5. Write sample code to embed both in a webpage.
10. Explain any five CSS properties used for layout design and styling.
11. Define Client Side and server side architecture. What is the difference between them?

Section C – Long Answer Questions (2 × 7.5 = 15 Marks)

(Attempt any 2 questions)

12. Create a web page layout using Flexbox that includes the following sections: Header, Left Sidebar, Main Content, and Right Sidebar. Write the HTML and CSS code with layout diagram.
13. Discuss JavaScript variable types, operators, and control structures with examples.
14. **OR**
    What is the Domain Name System (DNS)? Explain how domain name registration and web hosting work together to display a website online.

The End *(Good Luck)*

Gandaki University
Pokhara

BIT /Midterm Examination /2082
Course: Web Technology I (Sem-I)

Time: 2 Hrs.

Full Marks: 50
Pass Marks: 22.5

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full mark

Attempt all the questions:

Answer the following question in brief (Any five): **5x2=10**

1. Define URL. Write an example of a complete URL.
2. Write any two syntax rules of XHTML.
3. Write the full form of FTP and POP. State their basic function.
4. Write the difference between class and ID selectors in CSS with example.
5. List any two form input types and their uses.
6. What is a loop in JavaScript? Name two types of loops.

Answer the following questions (Any five): **5x5=25**

7. Explain HTTP, HTTPS, FTP, and SMTP protocols with appropriate use cases.
8. Explain with code:
   1. Ordered List 2. Hyperlink 3. Audio & Video 4. Image Insertion 5. Text Formatting
9. Explain the Box Model in CSS. Describe margin, border, padding, and content areas with a labeled diagram.
10. Define Client Side and server-side architecture. What is the difference between them?
11. Write a JavaScript function that accepts a number and checks whether it is even or odd. Use appropriate conditional and comparison operators.
12. Explain the use of internal and external CSS with examples.

Answer the following questions (Any two): **2x7.5=15**

13. Discuss the Domain Name System (DNS) and the hierarchy of domain names.
    Also explain the steps involved in domain name registration and compare types of web hosting (shared, VPS, dedicated).
14. Explain the purpose of HTML forms.
    Create a form with the following inputs:

- Username (text)
- Email (Email)
- Password (password)
- Gender (radio)
- File upload
- Courses (select, Option)
- Submit button

15 Design a responsive webpage layout using Flexbox. Include:

- A fixed header
- Three-column main content
- A footer

Explain:

- Flex properties used
- How responsiveness is achieved on mobile

The End

Gandaki University
Pokhara

BIT/Pre-Board Examination/2082 | Full Marks: 50
Course: Web Technology I/ Semester II | Time: 2 Hrs. | Pass Marks: 22.5

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full mark

Attempt all the questions:

Group A

Answer the following question in brief (Any five): 5x2=10

1. What is the function of a web server in web communication?
2. List the distinctions between FTP and SMTP.
3. Write the syntax to insert an image and hyperlink in HTML.
4. Explain how internal and external CSS differ from one another.
5. Define variable in JavaScript with an example.
6. Discuss the distinction between a session and a cookie.

Group B

Answer the following questions (Any five): 5x5=25

7. Create an HTML table to display product data with rowspan and colspan attributes. Include at least 3 rows and 3 columns.
8. Explain the different levels where CSS can be applied in a webpage. Include code samples for inline, internal, and external styles.
9. Write a JavaScript program that prints numbers from 1 to 10 using a while loop and for loop. Also, WAP in JavaScript code that changes the text color of a paragraph when a button is clicked.
10. What are events in JavaScript? Explain onmouseover and onkeyup and onclick with examples.
11. What is database? Write PHP code to insert a record into a MySQL table named **books** with fields id, title, author.
12. Define array and function. Describe the basic rules of PHP and how it handles form data using $_POST.

Group C

Answer the following questions (Any two): 2x7.5=15

13. Create a form in HTML that accepts name, username, phone and password. Use JavaScript to validate that fields are not empty and at least 5 characters long password, 10 digit phone number.
14. Consider a database named **web_portal**. Inside it, there is a table named **users** with the following fields:
    - id (INT, AUTO_INCREMENT, PRIMARY KEY)
    - name (VARCHAR)

- email (VARCHAR)

a) Write a PHP script to connect to the database and create table.
b) Write PHP script to fetch data by **SELECT** query, delete data by **DELETE** query and update all records by **UPDATE** query.
c) Explain briefly what are the advantages of PHP.

15 Write short Notes:
1. GET and POST method 2. CSS Layout techniques 3. Types of web hosting 4. Control statements in PHP with example 5. Types of variable scope in JavaScript.

The End

Gandaki University
Pokhara

BIT/Second Semester/2082/Final Examination — Full Marks: 50
CIT123 Web Technology- I — Time: 2 hrs — Pass Mark: 22.5

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all the questions.

1. Answer the following questions in brief (*any five*). **[5 × 2 = 10]**

a. Define a web browser and a web server.
b. Write the difference between HTTP and HTTPS.
c. Write the difference between and tag.
d. What are pseudo-classes in CSS? Provide two examples of commonly used pseudo-classes.
e. Write two differences between client-side and server-side scripting.
f. Define cookies.

Answer the following questions *(any five)*. **[5 × 5 = 25]**

2. Write the HTML code to generate the following table.

A test table with merged cells

	Average		Red eyes
	height	weight	
Males	1.9	0.003	40%
Females	1.7	0.002	43%

3. Write the HTML code to generate the following list.

1. Coffee
2. Tea
   - Milk Tea
   - Black tea
     a. Green tea
     b. Lemon Tea
3. Milk

4. Explain different types of CSS selectors with suitable example code.

5. What is the CSS box model? Explain the different components of the box model.

6. How does event handling work in JavaScript? Provide an example of using an event listener to trigger a function when a button is clicked.

7. What are session variables in PHP? Provide an example of setting and retrieving a session variable.

Answer the following questions *(any two)*. [2 × 7.5 = 15]

8. Compare and contrast the inline, internal, and external methods of embedding CSS in HTML. Also, write the advantages and disadvantages of each approach.
9. Write a Javascript program which validates user signup data (name, email, phone, password and re-password) and displays the success message if validated successfully. Also create an HTML form for entering the data.
10. Write a PHP code that connects with the database named "gandaki_university" and insert (name, email, age) in a table named "student".

The End

Gandaki University
Pokhara

BIT/Midterm Examination/2082
Course: Digital Logic (II Semester)

Time: 2 Hrs.

Full Marks: 50
Pass Marks: 22.5

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full mark

Attempt all the questions:

Answer the following question in brief (Any five): 5x2=10

1 What is signal ?

2 What is Clock signal ?

3 What is Numbering system.

4 Define ASCCII code.

5 What are Boolean's Theorem ?

6 What is don't care and explain why it is used in K-map?

Answer the following questions (Any five): 5x5=25

7 List out the differences between Analog System and Digital System.

8 Define Analog and Digital Signal. Which and why it is preferred in modern day of communication

9 What are Exclusive gates? Briefly explain the operation of Exclusive gates with truth table.

10 Perform the 1's Complement
   i) 10001
   ii) 110000

11 Briefly explain the FIRST and SECOND law of DEMORGAN'S Theorem.

12 Why NAND, NOR gate is called universal gate? Implement NOR gate as universal gate.

Answer the following questions (Any two): 2x7.5=15

13 Simplify the following expression using Boolean algebra
   i) X'Y +Y [Z'(Z'+Y)]
   ii) AB+(AC)'+AB'C (AB+C)

14 Perform the following conversion.
   i) $(35.75)_{10} = (\quad)_5$
   ii) $(64.23)_{16} = (\quad)_8$

iii) $(2643)_{10} = (\quad)_{\text{Excess-3}}$

iv) $(10101111)_{\text{Gray}} = (\quad)_{\text{Binary}}$

5 Simplify using K-map $F = \sum (1,3,7,11,15)$ and don't care condition $d = \sum (0,2,5)$ Implement the reduced expression using Basic Gate .

The End

Gandaki University
Pokhara

BIT/Assessment Examination
Course: Digital Logic (ELX 121)

Time: 2 Hrs.

Full Marks: 50
Pass Marks: 22.5

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full mark

Attempt all the questions:

Answer the following question in brief (Any five): **5x2=10**

1 Explain what is digital system?

2 Why do you prefer digital signal over analog?

3 What is Universal gate?

4 What is Gray Code.

5 Explain PLA.

6 What Read only Memory (ROM).

Answer the following questions (Any five): **5x5=25**

7 Perform the following conversion.

i) $(325)_{10} = (\quad)_7$

ii) $(253)_{10} = (\quad)_{Excess-3}$

iii) $(5432)_8 = (\quad)_{16}$

iv) $(1532)_7 = (\quad)_{10}$

v) $(645)_x = (962)_{10}$ Find the value of x.

8 Find 2's Complement

i) 10001

ii) 110111

9 Define Full adder combinational circuit. Show the full adder circuit is combination of two half adder and an OR gate.

10 What do you mean by Decoder? Implement the following Boolean function using 4*1 Mux $F=\sum(1,3,5,6)$

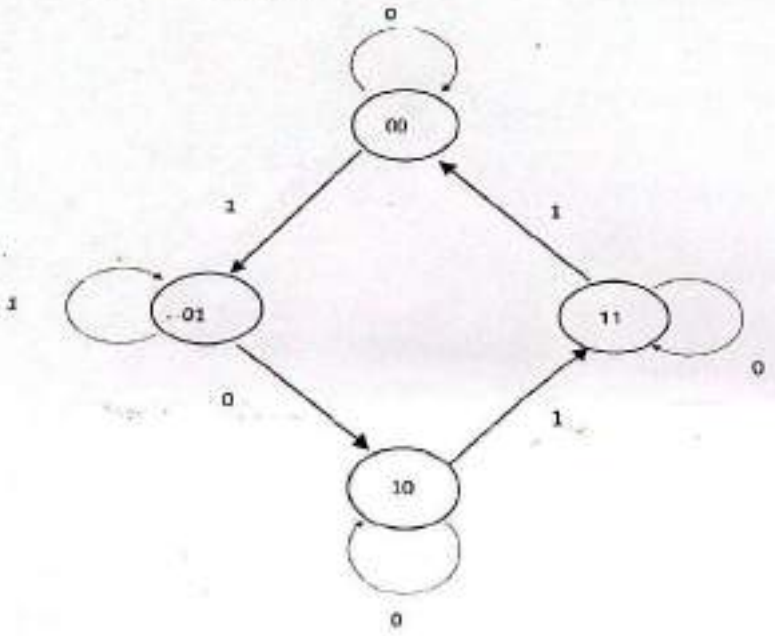
11 Explain operation of J-K Flip-flop with its logic diagram and truth table.

12 What is Counter and explain its type in brief.

Answer the following questions (Any two): **2x7.5=15**

13 A Boolean function is given by $F(A,B,C,D) = \sum(1,4,5,6,12,14,15)$ and don't care condition $d(A,B,C,D) = \sum(10,11)$. Simplify it using K-Map and implement using basic gate only.

14 Design a sequential circuit corresponding to the given state diagram using T Flip-Flop for the following state diagram.



15 What is Shift Register? Explain Parallel in Serial Out Shift Register (PISO) with its timing diagram.

Gandaki University
Pokhara

BIT/Second Semester/2082/Final Examination
ELX121 Digital Logic

Time: 2 hrs

Full Marks: 50
Pass Mark: 22.5

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all the questions.

1. Answer the following questions in brief *(any five).* **[5 × 2 = 10]**

a. Define analog and digital system. List the advantages of digital system over analog system.

b. List the application of digital system?

c. Why Nand, Nor gates are called Universal gate?

d. Why do we use don't care condition in K-Map?

e. Briefly explain the Encoder.

f. What is Asynchronous counter (Ripple Counter)?

Answer the following questions *(any five).* **[5 × 5 = 25]**

2. Perform the following conversion.

a. $(634)_{10} = (\quad)_5$

b. $(645)_{10} = (\quad)_{Excess\text{-}3}$

c. $(8AD4)_{16} = (\quad)_8$

d. $(345)_6 = (\quad)_{10}$

e. $(781)_x = (87)_{10}$ Find the value of x.

3. Find 1's Complement

a. 111001

b. 10001011

4. Implement and explain the half adder with its necessary logic diagram.

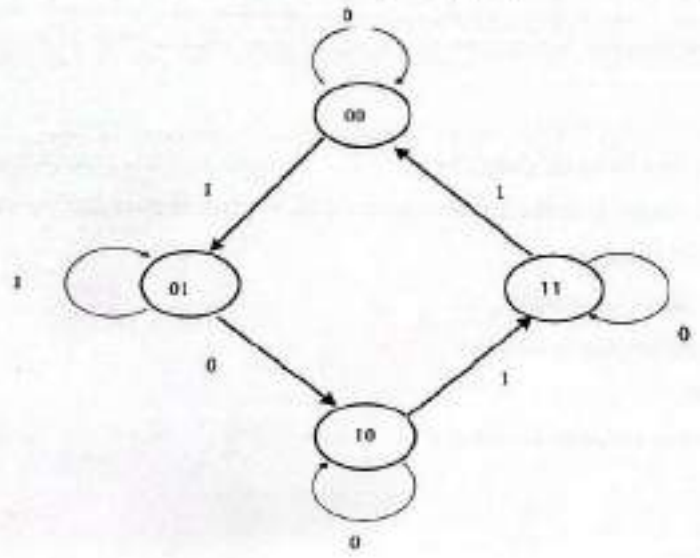
5. What do you mean by Demultiplexer? Implement the following Boolean function using 4*1 Mux $F = \sum(0,1,3,4,8,9,15)$.

6. Explain operation of J-K Flip-flop with its logic diagram and truth table.

7. Define Register. Explain the SISO with its working and timing diagram.

Answer the following questions *(any two)*. **[2 × 7.5 = 15]**

8. A Boolean function is given by F(A,B,C,D) = ∑(1,3,7,11,15) and don't care condition d(A,B,C,D) = ∑0,2,5). Simplify it using K-Map and implement using basic gate only.
9. Design a sequential circuit using D Flip-flop for the following state diagram.



10. Prove that the Full Adder is the combination of two half adder and an OR gate.

The End

Gandaki University
Pokhara
First Term Exam
2025

BIT Second Semester
Mathematics II (BSM 121)

Full Marks: 50
Time: 2 hour

Candidates are required to give their answer in their own words as far as practicable.
Figures in the margin indicate full marks.

Group A
Brief Answer Questions

Attempt any five questions [5 × 2 = 10]

1. How many numbers of three different digits can be formed from integers 1, 2, 3, 4, 5, 6?
2. If $C(n, 2) = C(n, 3)$ then find the value of ${}^{n}C_4$
3. Represent this $z = 4\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right)$ in rectangular form.
4. Test the analyticity of the function $f(z) = e^{-x}(\cos y - i\sin y)$.
5. By direct comparison test determine if the series $\sum_{n=1}^{\infty} 3^{-n}8^{n+1}$ is converge or diverge?
6. Using divergence test to show that the series $\sum_{n=1}^{\infty} \frac{n^2}{3n^2+4}$ diverges.

Group B
Short Answer Questions

Attempt any five questions. [5 × 5 = 25]

7. (a) In how many ways can 8 different colored beads be made into a bracelet?
   (b) How many distinct permutations can be formed using the letters of the word BIBLIOGRAPHY
8. In how many ways can a committee of 15 members formed out of 8 engineers, 5 businessmen and 4 lawyers so that all the lawyers always have a representation.
9. (a) Write the given complex number in the form of $a + ib$: $\frac{2-2i}{3+4i}$
   (b) Write the following complex number into polar form $z = 3 - 3i$
10. If $z = x + iy$ then use trigonometric and hyper trigonometric identities to show $\sin z = \sin x \cosh y + i\cos x \sinh y$. Also prove that $Re(\sin z)$ and $Im(\sin z)$ are harmonic.
11. State the Condition for convergence and divergence of infinite series using ratio test. Perform the convergence test on $\sum_{n=0}^{\infty} \frac{(-10)^n}{4^{2n+1}(n+1)}$

12. Write the standard Taylor's series expansion of $f(z)$ about $z = z_0$. Then determine the Maclurin series of $\sin z$.

Group C
Long Answer Questions

Attempt any **two** questions. $[2 \times 7.5 = 15]$

13. Consider the series $\sum_{n=0}^{\infty} \frac{n}{(n+1)!}$. If S_n denotes the partial sums of first n terms.

    (a) Find the partial sums S_1, S_2, S_3, S_4. Do you recognize the denominators? Use the patterns to find a formula for S_n

    (b) Show that the given infinite series is convergent and find its sum.

14. Define a p-series. State the condition for the convergence and divergence by using p-series. The harmonic series is also called p-series, why?

    Find the center, radius and interval of convergence of the power series $\sum_{n=0}^{\infty} \frac{(2n)!}{n!}(z-2i)^n$

15. Write the Cauchy-Riemann conditions for analyticity and Laplace's equation for harmonicness of the complex variable function. Is the functions $u = x^2 - y^2$ harmonic? If yes, find a harmonic conjugate and corresponding analytic function $f(z) = u(x, y) + iv(x, y)$.

The End

:7.5=15

Gandaki University
Pokhara
Pre-Board Exam
2025

BIT Second Semester
Mathematics II (BSM 121)

Full Marks: 75
Time: 3 hours

Candidates are required to give their answer in their own words as far as practicable.
Figures in the margin indicate full marks.

Group A
Very Short Answer Questions

Attempt any ten questions $10 \times 2 = 20$

1. How many numbers of different digits between 4000 and 5000 can be formed with the integer 1,2,3,4,5,6,7?
2. Compute: $\frac{(n+1)!}{n!}$.
3. Using divergence test show that the series $\sum_{n=1}^{\infty} \frac{n}{2n-5}$ diverges.
4. Test the convergence and divergence of the series: $1 + \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} + \frac{1}{\sqrt{4}} \cdots$
5. If $f(x, y, z) = x^2y + y^2z + xz^2$ find f_x, f_y, f_z
6. Verify the mixed derivative theroem (Euler's Theorem) for a function $z = x \sin y + y \sin x + xy$
7. Define the order and degree of a differential equation. What is the order and degree of the differential equation: $\left(\frac{d^3y}{dx^3}\right)^2 + \frac{d^2y}{dx^2} + \frac{dy}{dx} = 5x^2$
8. Determine whether the given equation is exact or not. $\cos(x+y)dx + (3y^2 + 2y + \cos(x+y))\,dy = 0$
9. Find the general solution of the differential equation $\frac{dy}{dx} = \frac{-2x}{y^2}$
10. Express $z = 3 + 3\sqrt{3}i$ into polar form.
11. If $z_1 = 3 - 7i$ and $z_2 = 2 + 3i$, then verify $\overline{(z_1 z_2)} = \overline{z_1}\,\overline{z_2}$

Group B

Attempt any **Eight** questions $8 \times 5 = 40$

12. A committee of 5 members is to be chosen from 9 ladies and 8 men. In how many ways can this be done if the committee commands a lady majority?

13. Determine whether the following geometric series $\sum_{n=1}^{\infty} \frac{(-3)^{n+1}}{4^{n-1}}$ converges or diverges, and if it converges, find its sum.

14. For the given series $\sum_{n=0}^{\infty} \frac{1}{2^n - 1}$
    (a) Show that the convergence and divergence test fails by using comparison test.
    (b) Use limit comparison test to prove that the series converges.

15. Find the total derivative $\frac{dz}{dt}$ and $\frac{d^2z}{dt^2}$ of the function $z = 4x^2 + 3y^2$, $x = \sin t$, $y = \cos t$

16. Find all the critical points and classify each as a relative maximum, a relative minimum, or a saddle point of the given functions: $f(x, y) = x^3 + y^3 - 4xy + 12$

17. Solve the initial value problem $y'' + y = 0$, $y(0) = 3$, $y'(0) = -0.5$

18. Find the general and particular solutions of the given linear differential equation: $\frac{dy}{dx} + \frac{y}{x} = x^2$, $y(1) = 1$

19. (a) Express the complex number $z = \frac{2 - 3i}{4 + 3i}$ in the form of $a + ib$
    (b) Find the square root of $5 - 12i$

20. State the De'Moivre theorem, Use it to find the cube root of unity.

Group C

Attempt any **Two** questions $2 \times 7.5 = 15$

21. State the condition for the convergence and divergence of the series using ratio test. Find the center, radius and interval of convergence of the power series

$$\sum_{n=1}^{\infty} \frac{n}{2^n} (z + i)^{2n}$$

22. Define an analytic function of complex variable. Verify that $u = x^3 - 3xy^2$ is harmonic in the whole complex plane and find a harmonic conjugate function $v(x, y)$ of $u(x, y)$ also find the analytic function.

23. Define the exact differential equation. Determine whether the given differential equation $(x^2 + y^2)dx - 2xydy = 0$ is exact or not? then solve it.

The End

Gandaki University
Pokhara

BIT/Second Semester/2082/Final Examination — Full Marks: 75
BSM121 Mathematics II — Time: 3 hrs — Pass Mark: 34

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all the questions.

1. Answer the following questions in brief (*any ten*). **[10 × 2 = 20]**

a. If C(8, r-1) = C(8, 3), find r.

b. State the conditions for the convergence and divergence of an infinite series by p - Test.

c. Find f_x and f_y if f(x, y) = $ln\sqrt{x^3 + 2y^2}$

d. Define integrating factor of differential equations. Find the integrating factor of the differential equation $cosx\,\frac{dy}{dx} + y\sin x = 1$.

e. Find the real and imaginary parts of the complex number e^{iz}.

f. In how many ways can 8 students be seated around a table?

g. Verify the mixed derivative theorem on $F(x,y) = ax^2 + 2hxy + by^2$.

h. Solve: x dy = y dx

i. State the Cauchy-Riemann conditions for analytic function $F(z)$. Is $z = e^{2x}(\cos y + i\sin y)$ analytic?

j. Check the exactness of the differential equation $(x^3 + 3xy^2)dx + (3x^2y + y^3)dy = 0$

k. Express the complex number $\frac{2-\sqrt{-36}}{1-\sqrt{-16}}$ in the form a + i b.

Answer the following questions *(any eight)*. **[8 × 5 = 40]**

2. A student is required to answer 6 out of 10 questions which are divided into 2 groups each containing 5 questions and he is not permitted to attempt more than 4 from any group. In how many different ways can he make up his choice?

3. Given $z = e^x.tany$, find $\frac{\delta^2 z}{\delta x \delta y}$ and $\frac{\delta^2 z}{\delta y \delta x}$.

4. Determine the location and nature of all the critical points of the function
f(x, y) = $3x^2 - xy + 2y^2 - 4x - 7y + 12$

5. Solve the initial value problem.
$y'' - 2y' - 3y = 0, \quad y(0) = 2, \quad y'(0) = 14$

6. Use De' Moivre's theorem to solve $z^3 = 1$

7. Discuss the convergence and divergence of the series:
$$\sum_{n=1}^{\infty} \frac{9^n}{(-2)^{n+1}.n}$$

8. If $u = \ln(x^3 + y^3 + z^3 - 3xyz)$, then prove that $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = \frac{3}{x+y+z}$

9. Solve the given differential equation: $2xy\frac{dy}{dx} = y^2 - x^2$

10. Check for analyticity of

    a. $f(z) = z^2$ b. $f(z) = \bar{z}$

Answer the following questions *(any two)*. **[2 × 7.5 = 15]**

11. Find the centre, radius of convergence and interval of convergence of the series
$x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots \ldots \ldots \ldots \ldots$

12. Prove that $u = y^3 - 3x^2y$ is a harmonic function. Determine its harmonic conjugate and find the corresponding analytic function f(z) in terms of z. **[2 + 4 + 1.5]**

13.

    a. If $F(x, y)$ is a homogeneous function of degree n, i.e $F(x, y) = x^n . F(\frac{y}{x})$, then prove that $x\frac{\partial f}{\partial x} + y\frac{\partial f}{\partial y} = nf$

    b. Solve the differential equation: Solve: $\frac{dy}{dx} + \frac{y}{x} = 4x^2$ **[4 + 3.5]**

The End

Gandaki University
Pokhara

BIT /Midterm Examination /2082 | Full Marks: 50
Course: Programming II (Java) | Time: 2 Hrs. | Pass Marks: 22.5

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full mark

Attempt all the questions:

Answer the following question in brief (Any five): 5x2=10

1 Why do we use abstraction in OOP?

2 What is JIT and why is it used in Java?

3 Define Class and Object with examples.

4 What is an array and how do you create a 2D array in Java?

5 What is an infinite loop? Give an example using while loop.

6 What is the purpose of a package in Java?

Answer the following questions (Any five): 5x5=25

7 Differentiate between Object-Oriented Programming and Procedural Programming.

8 Why is Java called platform-independent and how does Java achieve it?

9 Explain data types in Java with examples.

10 Write a Java program to demonstrate the use of bitwise and bitwise shift operators.

11 Why do we perform debugging? Describe some common debugging techniques used in Java.

12 Write a Java program to demonstrate inheritance. The program should meet the following requirements:

- Create a base class Employee with method displaySalary().
- Create a subclass Manager that adds a method manageTeam().
- Create a Manager object and call both methods.

See Convert human readable code to machine bitwise and vice versa

Answer the following questions (Any two): 2x7.5=15

13 How can you implement and apply interface to a class? Explain with code example.

14 Write a Java program that reads two integers from the user and performs a division operation (first number divided by second). Implement appropriate exception handling to manage the following exceptions:

- InputMismatchException (when user inputs non-integer values)
- ArithmeticExceeption (e.g., division by zero).

15 Explain how serialization and descrialization process works in Java.

The End

Gandaki University
Pokhara

BIT/Pre-Board Examination/2082
Course: Programming II (Java)/ Sem II

Time: 2 Hrs.

Full Marks: 50
Pass Marks: 22.5

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full mark

Attempt all the questions:

5x2=10

Answer the following question in brief (Any five):

1 What is a constant in Java? How is it declared?
2 How many times will the following statement execute and what will it print?
```
int x = 5;
String result = (x % 2 == 0) ? "Even" : "Odd";
System.out.println(result);
```
3 What is the purpose of the finally block in exception handling?
4 Write two differences between byte streams and character streams.
5 Name any two event listener interfaces.
6 Point out two disadvantages of multicasting.

5x5=25

Answer the following questions (Any five):

7 Explain any four key concepts of Object-Oriented Programming with suitable examples.
8 Explain the process of how code is compiled and executed in Java.
9 Explain the structure of a basic Java program. Highlight the role of the main() method in program execution.
10 Write a Java program that demonstrates the use of break and continue in a loop. Explain how control flow is affected.
11 Describe the hierarchy of Exception classes in Java. Mention the difference between checked and unchecked exceptions.
12 Write a program to open a file, write integers at specific positions using seek(), and then read and print them.

2x7.5=15

Answer the following questions (Any two):

13 Explain access protection in Java and discuss its effect on inheritance. Write a Java program to demonstrate how protected and private members behave when a class is extended.
14 Explain the Delegation Event Model in Java. Write a program to demonstrate handling a button click using the ActionListener interface.
15 Compare and contrast AWT and Swing with examples. Discuss the use of MVC pattern in context of Swing.

The End

Gandaki University
Pokhara

BIT/Second Semester/2082/Final Examination — Full Marks: 50
CIT122 Computer Programming II (Java) — Time: 2 hrs — Pass Mark: 22.5

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all the questions.

1. Answer the following questions in brief (*any five*). **[5 × 2 = 10]**

a. How is object-oriented programming different from procedure-oriented programming?
b. What are the three main components of the Java Development Kit?
c. Define variables and constraints. Provide an example of each.
d. What is the difference between error and exception?
e. What are Abstract Classes in Java and why are they important?
f. What do you mean by Layout Management?

Answer the following questions *(any five)*. **[5 × 5 = 25]**

2. Differentiate between procedural oriented programming and object-oriented programming.
3. What makes Java platform independent? Explain.
4. List out the different types of operators in Java. Explain how for loop helps in decision making in Java.
5. What are exceptions? What are the different types of exceptions in Java and how will you handle them?
6. Explain how to read from and write to a file in Java with an example program.
7. What do you mean by event-handling? Describe Event Delegation Model in Java.

Answer the following questions *(any two)*. **[2 × 7.5 = 15]**

8. Write a Java program to get 10 array elements from the user and print only the odd numbers. **Explain public static void** main(String[] **args**){}.
9. What are the core principles of Object Oriented programming? List out all the types of inheritance with respective examples.
10. Compare AWT with Swing. Create a simple Swing application with a JFrame containing a JLable displaying "Hello Swing!".

The End

Gandaki University
Pokhara

IT/ Practical Examination /2082
ourse: Computer Programming – II
ava)

SET - A
Time: 30 min

Full Marks: 10
Pass Marks: 4

empt all the questions:

1. Create a Java program based on the following criteria:
   i. Create an interface named *VehicleInfo* with the following methods: **inputDetails, displayDetails.**
   ii. Create a base class named *Vehicle* with the following data members: **brand, model, year** and initialize using constructor.
   iii. Create a derived class Car that inherits from Vehicle, Implements the *VehicleInfo* interface, includes an additional data member *fuelType*, Implements the **inputDetails** and **displayDetails** methods
   iv. Create the *Car* object, Input the car details, and display the car details using the interface.

The End

Gandaki University
Pokhara

BIT /Midterm Examination /2082 | Full Marks: 50
Course: Discrete Mathematics (Sem-II) | Time: 2 Hrs. | Pass Marks: 22.5

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full mark

Attempt all the questions:

Answer the following question in brief (Any five): 5x2=10

1. Define atomic and compound proposition.
2. Define Tautology.
3. Explain Handshaking Theorem.
4. State De-Morgan's Law.
5. A simple graph G has 24 edges, with the degree of each vertex 4. Find the number of vertices.
6. Define the Bipartite graph.

Answer the following questions (Any five): 5x5=25

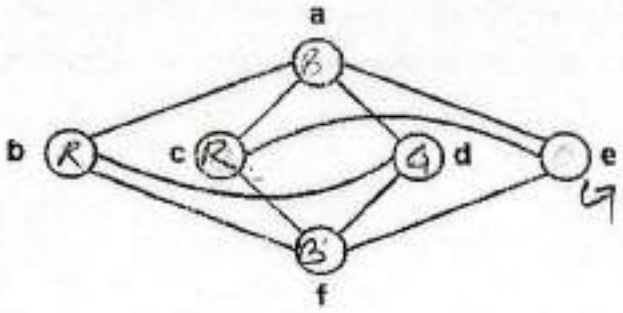
7. What are logical connectives? Discuss If Then, If and Only If connective along with their truth table.
8. Using direct proof, prove that for every positive integer $n^3 + n$ is even.
9. Prove $\neg(p \vee (\neg p \wedge q)) \equiv (\neg p \wedge \neg q)$ using logical equivalence.
10. Consider these statements:
    - No students are lazy.
    - All lazy people are procrastinators.
    - Some students are lazy.

    Express the above statement using quantifiers where the domain consists of all people.
11. Write any five rules of inference for propositional logic.
12. Show that the premises "It is not sunny this afternoon and it is colder than yesterday," "We will go swimming only if it is sunny," "If we do not go swimming, then we will take a canoe trip," and "If we take a canoe trip, then we will be home by sunset" lead to the conclusion "We will be home by sunset."

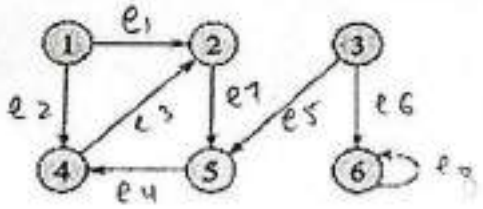
Answer the following questions (Any two): 2x7.5=15

13 Use Mathematical induction to show that: $8^n - 3^n$ is divisible by 5.[$n \geq 1$]

14 What do you mean by Graph Coloring and Chromatic number? Find the chromatic number for the following graph:



15 Write Adjacency Matrix, Incidence Matrix, and Adjacency List for the following graph.



The End

Simple

M T

M I

M P

B, C – R

Gandaki University
Pokhara

BIT/Pre-Board Examination /2082 Full Marks: 75
Course: Discrete Mathematics Time: 3 Hrs. Pass Marks: 34
(Second Semester)

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full mark

Group A

Brief Answer Questions: (Any Ten Questions) **10*2=20**

1. Define quantifiers.
2. What do you mean by logical equivalence?
3. Define Contingency.
4. Define Existential Instantiation.
5. Define FSM.
6. Find the characteristic roots of the recurrence relation $a_n = 3a_{n-1} - 3a_{n-2} + a_{n-3}$.
7. Find the recurrence relation and initial conditions of the sequence 3, 8, 13, 18, ...
8. Explain Isomorphic graphs.
9. Explain the Regular graph with an example.
10. Let G be a connected planar simple graph with 25 vertices and 60 edges. Find the number of regions in G.
11. Explain theorem "An undirected graph has an even number of vertices of odd degree".

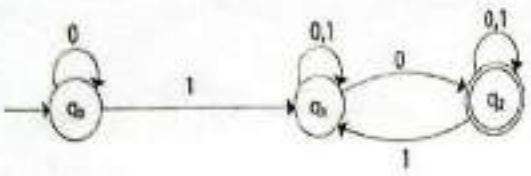
Group "B"

Short Answer Questions: (Any Eight Questions) **8^5=40**

1. Let p and q be the propositions: (Consider p and q yourself)
   Express each of these propositions as an English sentence:
   a) $\neg p$ b) $\neg p \wedge \neg q$ c) $\neg p \vee (p \wedge q)$ d) $p \leftrightarrow q$
2. State the converse, contrapositive, and inverse of each of these conditional statements:
   a. If it rains tonight, then I will watch movie.
   b. I go to the swimming pool whenever it is a sunny summer day.
3. Show that if n is positive integer then, $1 + 2 + \ldots\ldots + n = [n(n+1)]/2$.
4. Show that the premises "All rock music is loud", "Some rock music exist", imply the conclusion "Some Loud music exists".
5. Draw the transition diagram of the finite state machine M, where, I = {a, b} O = {0, 1} S = {P, Q, R} σ = P and transition gives by the below table. Find the output string corresponding to the input string "aabbaba".

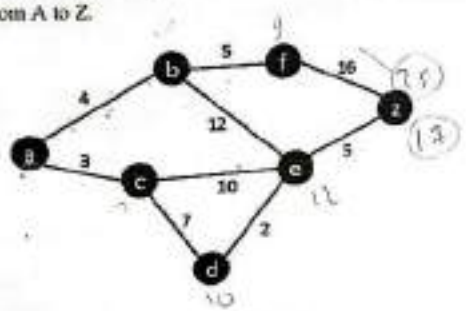
	σ		g	
S/I	a	b	a	b
P	Q	Q	0	1
Q	R	Q	1	1
R	P	P	0	0

6. Design a DFA that accepts the set of all strings ending with 101 over $\Sigma = \{0, 1\}$.
7. Convert the following NFA into an equivalent DFA.



8. Derive an Explicit Formula for the Fibonacci series.

9. Apply Dijkstra's algorithm to find the following graph's shortest path from A to Z.



Group "C"

Long Answer Questions (Any Two Questions) 2*7.5=15

1. Solve the recurrence relation $a_n - 5a_{n-1} + 6a_{n-2} = 2^n n$ with initial conditions $a_0=1$ and $a_1=1$.

Adjacency list.

vertex Adjacent vertices

1 →

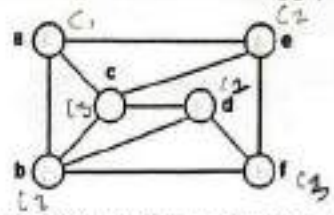
2 →

3 →

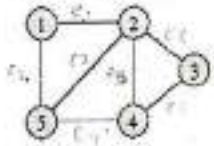
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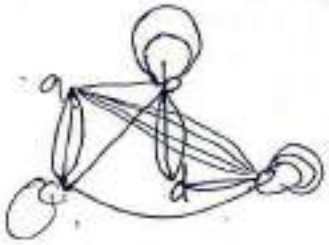
5

2. What do you mean by Graph Coloring and Chromatic number? Find the chromatic number for the following graph



3. Write Adjacency Matrix, Incidence Matrix, and Adjacency List for the following graph.





The End

Incident

	e_1	e_2	e_3	e_4	e_5	e_6	e_7
v_1	1	1	0	0	0	0	0
v_2	1	0	1	0	1	1	0
v_3	0	0	0	0	0	1	1
v_4	0	0	0	1	1	0	1
v_5	0	1	1	1	0	0	0

Gandaki University
Pokhara

BIT/Second Semester/2082/Final Examination — Full Marks: 75
CIT121 Discrete Mathematics — Time: 3 hrs — Pass Mark: 34

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

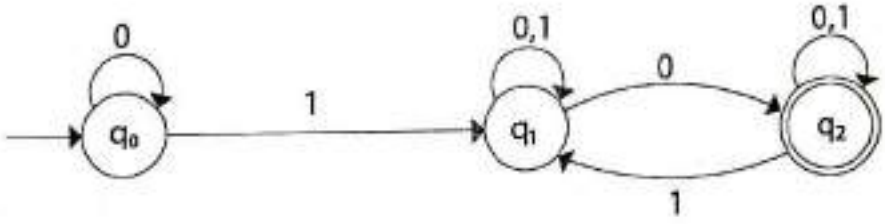
Attempt all the questions.

1. Very short questions *(Any ten).* **[10 × 2 = 20]**

a. Define quantifiers.
b. Find the negation of the statement "Some students are smart".
c. Briefly explain handshaking theorem.
d. What do you mean by logical equivalence?
e. Find the characteristic roots of the recurrence relation $a_n = 3a_{n-1} - 3a_{n-2} + a_{n-3}$.
f. Show that implication and its contrapositive are logically equivalent.
g. Define Tautology.
h. A simple graph G has 24 edges, with the degree of each vertex 4. Find the number of vertices.
i. Find the recurrence relation an initial condition of the sequence 3, 8, 13, 18,.....
j. When a relation is said to be homogeneous?
k. Define regular graph with example.

Short questions *(Any Eight).* **[8 × 5 = 40]**

2. Design a DFA with $\sum = \{0, 1\}$ accepts the set of all strings with three consecutive 0's.
3. Convert the following NFA into equivalent DFA.

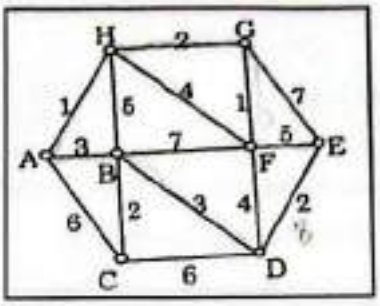


4. Use mathematical induction to show that: $8n - 3n$ is divisible by 5, [n>1].
5. Write any five rules of inference for propositional logic.
6. Hypothesis: "If Tuesday, I have a test in mathematics or English. If my English professor is sick, I will not have a test in English. Today is Tuesday and my English professor is sick. Therefore, I have a test in Mathematics". Show that the following argument is valid.
7. Prove $\neg(p \vee (\neg p \wedge q)) \equiv (\neg p \wedge \neg q)$ using logical equivalence.

8. Use mathematical induction to prove that $n^3 - n$ is divisible by 3 where n is a positive integer.

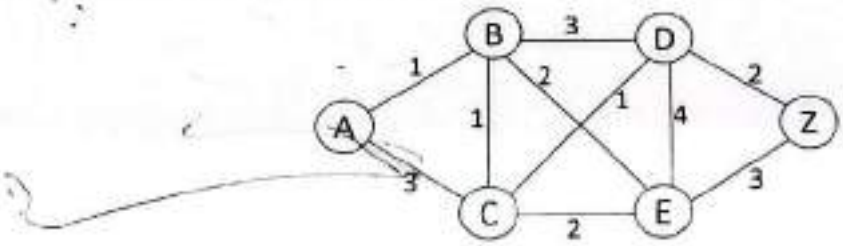
9. What are the real–life examples of graph? Explain the different graph representation techniques with suitable example.

10. Apply Dijkstra's algorithm to find the following graph's shortest path from A to E.

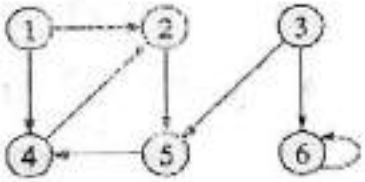


Long questions *(Any Two)*. **[2 × 7.5 = 15]**

11. Find all-to-all path from given graph.



12. Define Graph representation. Write Adjacency matrix, Incidence matrix and Adjacency List for the following graph.



13. Define linear homogeneous recursion of degree K with constant coefficient with suitable examples. What is the solution of the recurrence relation $a_n = a_{n-1} - 2a_{n-2}$ with $a_0 = 2$ and $a_1 = 7$.

The End